

UNIVERSITÀ DEGLI STUDI DI MILANO

ANNPR 2026

Counterfactual Explanations for Graph Neural Networks in Patient Outcome Prediction

N. Chaidos, A. Dimitriou, H. Calzi, E. Casiraghi, G. Stamou, and G. Valentini

AI|LS



Patient Similarity Networks (PSNs)

- **Nodes**: patients
- **Edges**: clinical and biomolecular similarities between patients

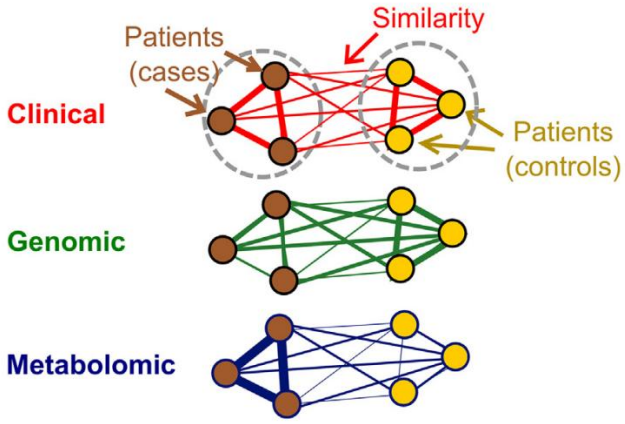


Figure: [Pai '18]

Patient Similarity Networks (PSNs)

- **Nodes**: patients
- **Edges**: clinical and biomolecular similarities between patients

Overcoming **clinical data challenges:**

- **Multi-modal integration**
  (e.g., via SNF [Wang '14])
- **Low sample cardinality**
  (leveraging graph topology)
- **Missing patient views**
  (recovered via miss-SNF [Gliozzo '25])

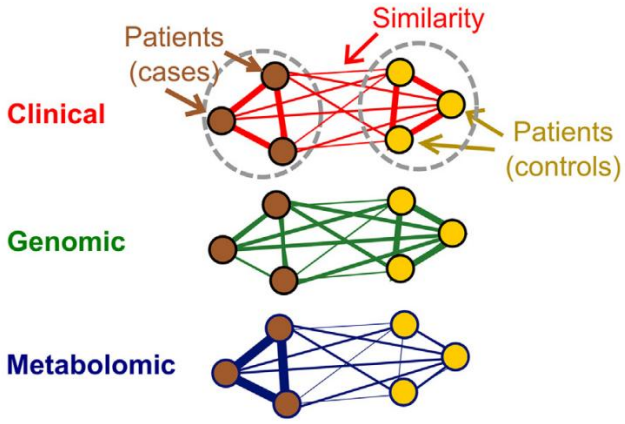


Figure: [Pai '18]

The Need for XAI in Clinical AI

- **Graph Neural Networks** (GNNs) have **predictive capability**
- GNN **black box** nature **limits** their adoption in the **clinical** setting
- Needs of transparency in decision-making

The Need for XAI in Clinical AI

- **Graph Neural Networks** (GNNs) have **predictive capability**
- GNN **black box** nature **limits** their adoption in the **clinical** setting
- Needs of transparency in decision-making



Explainable Artificial Intelligence (XAI)

Counterfactual Explanations (CE)

- **Explainable AI** technique, answers *what-if* questions
- Finds the **minimum perturbation** required to alter a model's prediction

"if your BMI were lower than 29, your predicted outcome would be *survived* instead of *deceased*"

Counterfactual Explanations (CE)

- **Explainable AI** technique, answers *what-if* questions
- Finds the **minimum perturbation** required to alter a model's prediction

"if your BMI were lower than 29, your predicted outcome would be *survived* instead of *deceased*"

Unlike classical feature selection methods, CE allows to:

- **Instance-level personalization:** Identifying disease drivers specific to the single patient
- **Actionability:** Direct therapeutic intervention

The Gap: Counterfactual Explainers on PSNs

- Tabular CE methods **ignore relational context** (DiCE [Mothilal '20], SingleCF [Wachter '18])
- Graph CE methods focus mostly on **graph classification** (CLEAR [Ma '22])
- Lack of multimodal **integration**

No existing CE algorithm is tailored for *node classification* on PSNs in clinical outcome prediction.

The Gap: Counterfactual Explainers on PSNs

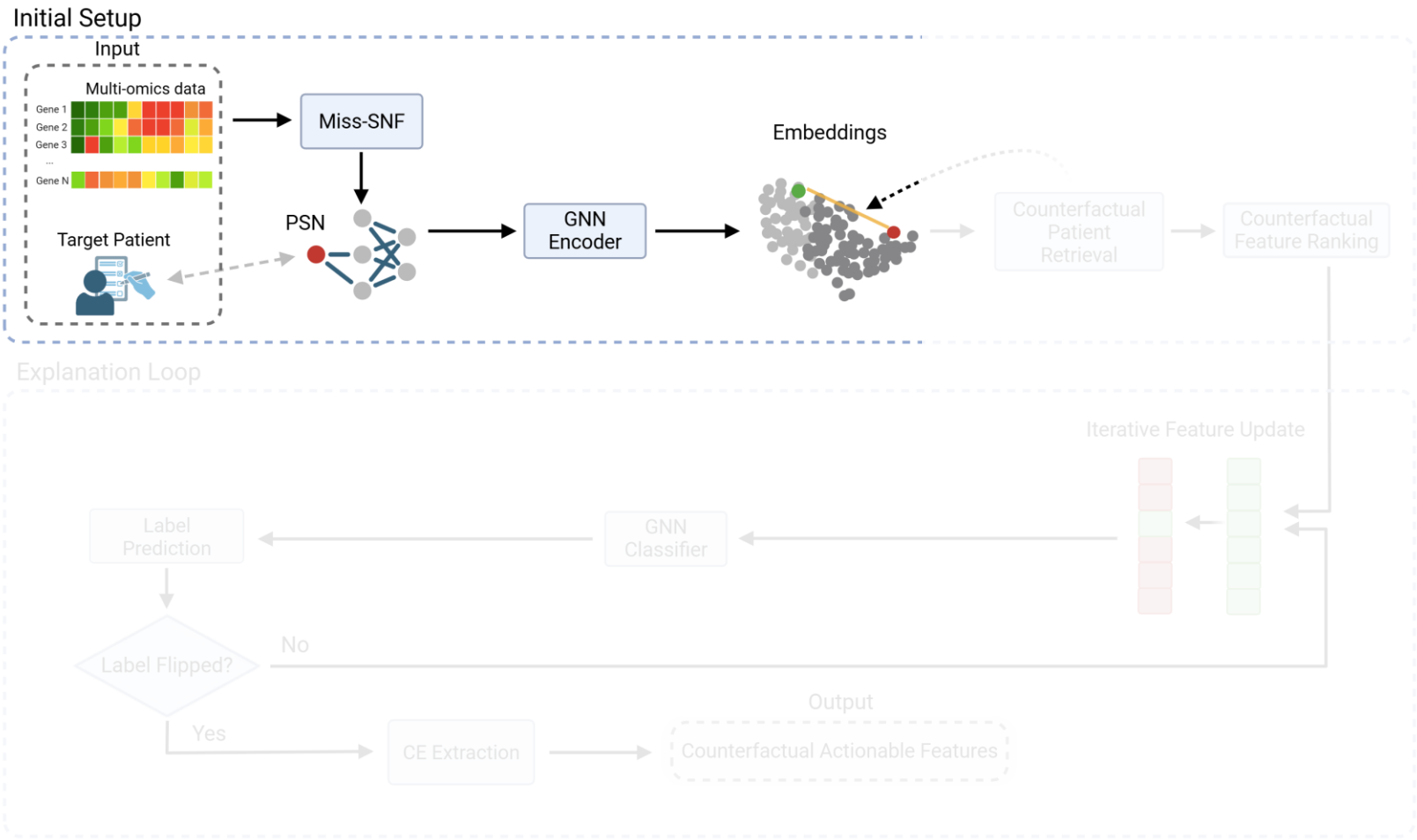
- Tabular CE methods **ignore relational context** (DiCE [Mothilal '20], SingleCF [Wachter '18])
- Graph CE methods focus mostly on **graph classification** (CLEAR [Ma '22])
- Lack of multimodal **integration**

No existing CE algorithm is tailored for *node classification* on PSNs in clinical outcome prediction.

Our **solution**: **C**ounterfactual **E**xplanations **G**raph **N**eural **N**etworks (**CE-GNN**)

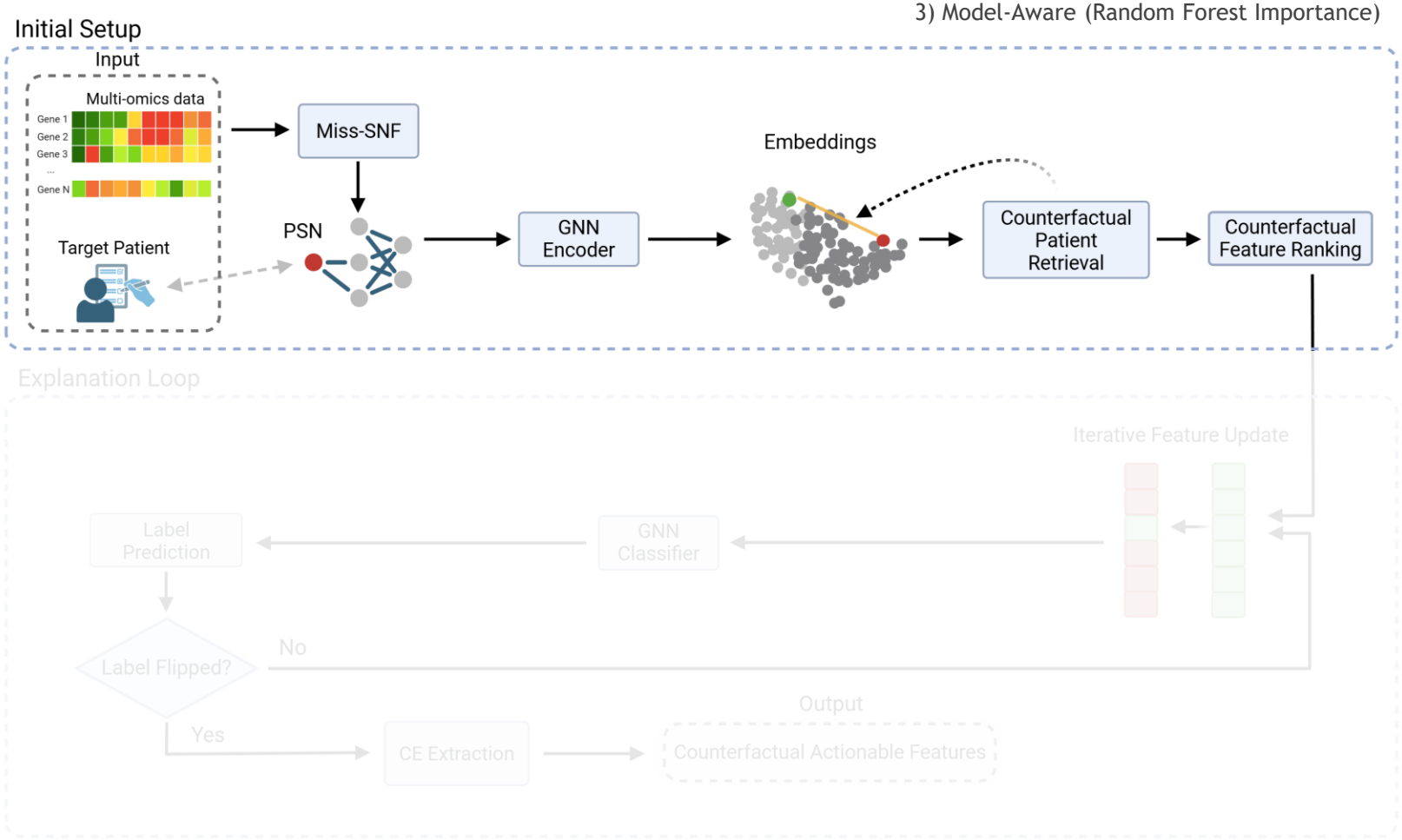
CE-GNN: Initial Setup

Contrastive Loss: pulls same-outcome patients together, pushes different outcomes apart

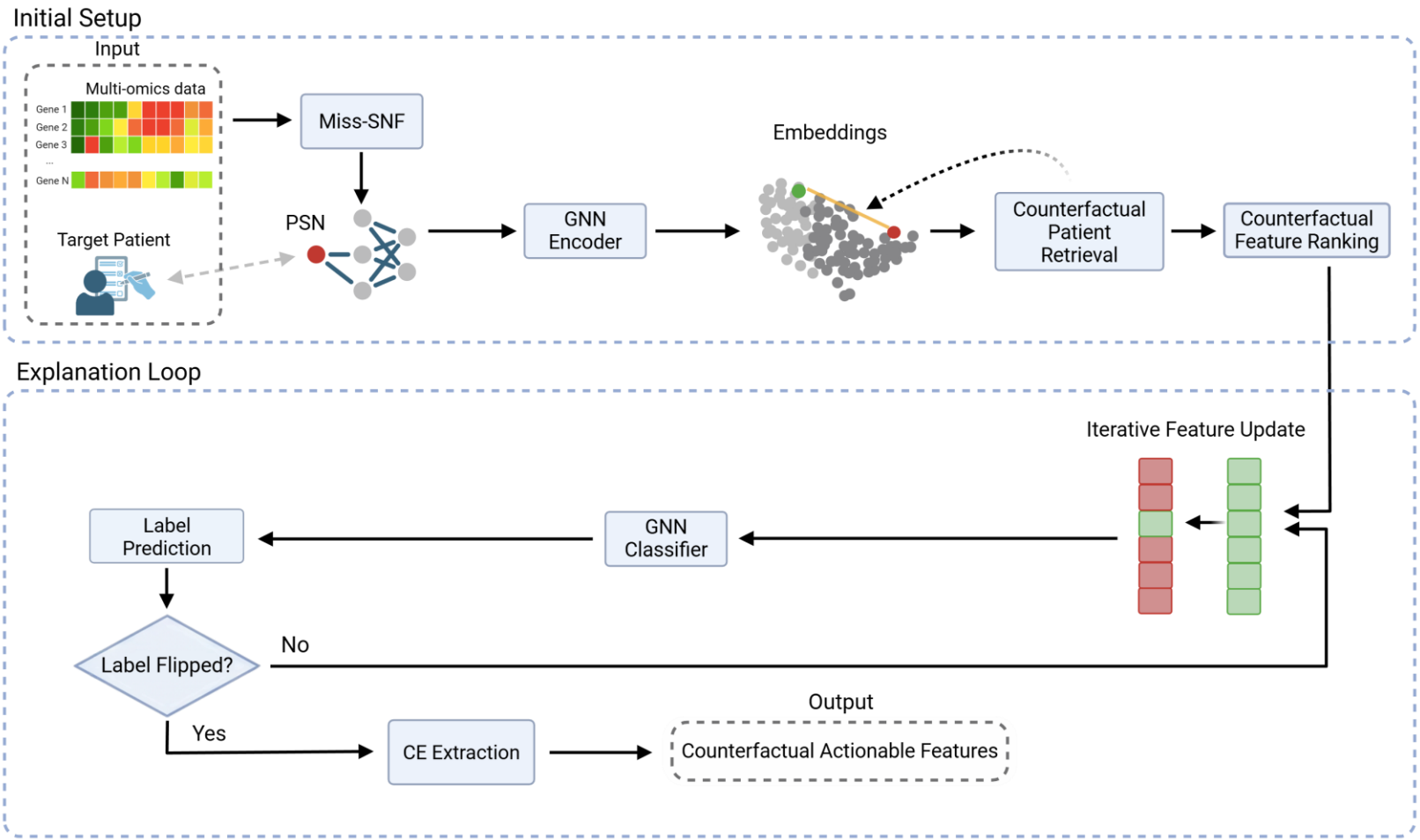


CE-GNN: Patient Retrieval & Feature Ranking

Ranking strategies: 1) Absolute Difference ($|f^i - f^{ci}|$),
2) Statistical Significance (ANOVA F-value),
3) Model-Aware (Random Forest Importance)



CE-GNN: Explanation Loop



Experimental Setup

- **Dataset:**
  - **Synthetic** Networked Data (with known ground truth)
  - **ROSMAP** (Alzheimer's Disease (AD) dataset, curated by MOGONET [Wang '21])
- **Proposed framework:**
  - **CE-GNN** with 3 ranking strategies: Absolute Difference (Abs), Statistical (St), Random Forest (RF)
- **Baselines:**
  - Counterfactual tabular-based: **DiCE** [Mothilal '20], **SingleCF** [Wachter '18]
  - Factual graph-based: **GNNExplainer** [Ying '19]
- **Metrics:**
  - **Ground truth** recovery: Precision, Recall, Macro-F1, Accuracy
  - **Quality** of generated **counterfactuals**: Validity, Fidelity, Sparsity

Experimental Setup

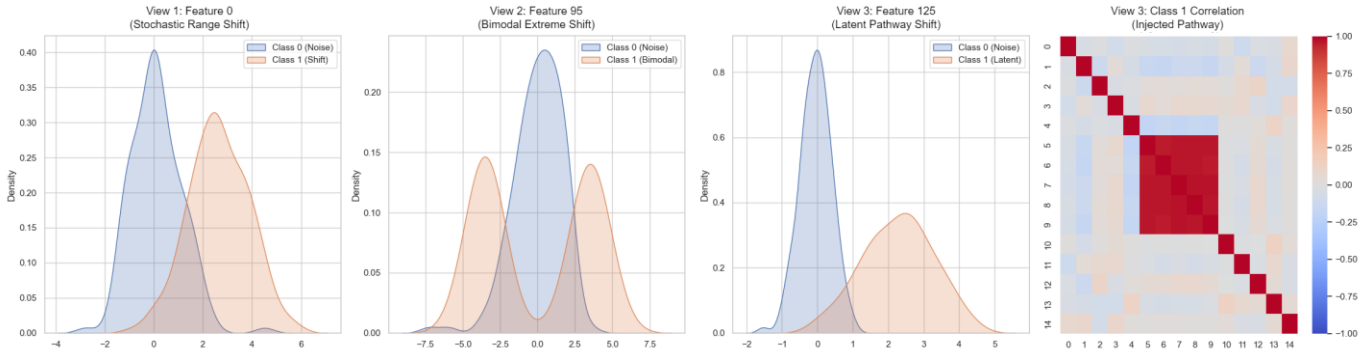
- Dataset:
  - **Synthetic** Networked Data (with known ground truth)
  - **ROSMAP** (Alzheimer's Disease (AD) dataset, curated by MOGONET [Wang '21])
- **Proposed framework:**
  - **CE-GNN** with 3 ranking strategies: Absolute Difference (Abs), Statistical (St), Random Forest (RF)
- **Baselines:**
  - Counterfactual tabular-based: **DiCE** [Mothilal '20], **SingleCF** [Wachter '18]
  - Factual graph-based: **GNNExplainer** [Ying '19]
- Metrics:
  - **Ground truth** recovery: Precision, Recall, Macro-F1, Accuracy
  - **Quality** of generated **counterfactuals**: Validity, Fidelity, Sparsity

Experimental Setup

- **Dataset:**
  - **Synthetic** Networked Data (with known ground truth)
  - **ROSMAP** (Alzheimer's Disease (AD) dataset, curated by MOGONET [Wang '21])
- **Proposed framework:**
  - **CE-GNN** with 3 ranking strategies: Absolute Difference (Abs), Statistical (St), Random Forest (RF)
- **Baselines:**
  - Counterfactual tabular-based: **DiCE** [Mothilal '20], **SingleCF** [Wachter '18]
  - Factual graph-based: **GNNExplainer** [Ying '19]
- **Metrics:**
  - **Ground truth** recovery: Precision, Recall, Macro-F1, Accuracy
  - **Quality** of generated **counterfactuals**: Validity, Fidelity, Sparsity

Validation on Synthetic Data

Method	Precision	Recall	macro-F1	Accuracy	Validity	Fidelity	Sparsity
GNNExplainer	57.31	35.89	44.14	**89.48**	-	-	**92.30**
SingleCF	10.54	**52.33**	17.55	51.44	**100.0**	**96.67**	50.98
DiCE	10.42	51.22	17.32	51.71	**100.0**	**96.67**	51.47
CE-GNN (Abs.)	11.94	37.11	18.07	73.76	98.33	95.00	76.17
CE-GNN (St.)	46.36	34.78	39.74	84.60	**100.0**	**96.67**	87.64
CE-GNN (RF)	**60.59**	38.22	**46.87**	86.73	**100.0**	**96.67**	89.09



Real-World Results: ROSMAP

Method	Validity	Fidelity	Sparsity
GNNExplainer	-	-	80.17
SingleCF	**100.0**	**70.75**	49.69
DiCE	**100.0**	**70.75**	51.24
CE-GNN (Abs.)	91.51	62.26	**85.75**
CE-GNN (St.)	85.85	56.60	75.69
CE-GNN (RF)	90.57	61.32	77.95

Real-World Results: ROSMAP

Method	Validity	Fidelity	Sparsity
GNNExplainer	-	-	80.17
SingleCF	**100.0**	**70.75**	49.69
DiCE	**100.0**	**70.75**	51.24
CE-GNN (Abs.)	91.51	62.26	**85.75**
CE-GNN (St.)	85.85	56.60	75.69
CE-GNN (RF)	90.57	61.32	77.95

Biological validation: most of the identified features align with known AD biomarkers in literature

Limitations & Future Work

Current **limitations**:

- Sub-optimal Validity on ROSMAP (< 100%)
- **Static topology** bottleneck (only node features are perturbed)

Limitations & Future Work

Current **limitations**:

- Sub-optimal Validity on ROSMAP (< 100%)
- **Static topology** bottleneck (only node features are perturbed)



Future ongoing **directions**:

- **Dynamic** CE-GNN
- Broader experimental benchmarking

Conclusions: CE-GNN

- **First** CE framework tailored for Patient Similarity Networks
- It is able to identify **actionable features** for **specific therapeutics** in a clinical setting
- Ongoing work on a **dynamic version** of CE-GNN perturbing both node features and network topology to achieve full validity

Thank you for your attention!

Hari Calzi - hari.calzi@unimi.it

AI|LS

Preprint